

Cost Concepts

A firm uses several inputs:

- Labour
- Machinery
- Raw materials
- Capital

All these inputs involve costs.

The basic reason we study cost is that a firm needs to decide:

1. How much should it produce?
2. What will production cost?
3. Will production be profitable?

Costs in the Short Run

- **Fixed cost** is any cost that does not depend on the firm's level of output. These costs are incurred even if the firm is producing nothing.
- **Variable cost** is a cost that depends on the level of production chosen.

$$TC = TFC + TVC$$

$$\begin{array}{ccccc} \text{Total Cost} & = & \text{Total Fixed} & + & \text{Total Variable} \\ & & \text{Cost} & & \text{Cost} \end{array}$$

Costs in the Short Run

- The *short run* is a period of time for which two conditions hold:
 1. The firm is operating under a fixed scale (fixed factor) of production, and
 2. Firms can neither enter nor exit an industry.
- In the short run, all firms have costs that they must bear regardless of their output. These kinds of costs are called *fixed costs*.

Fixed Costs

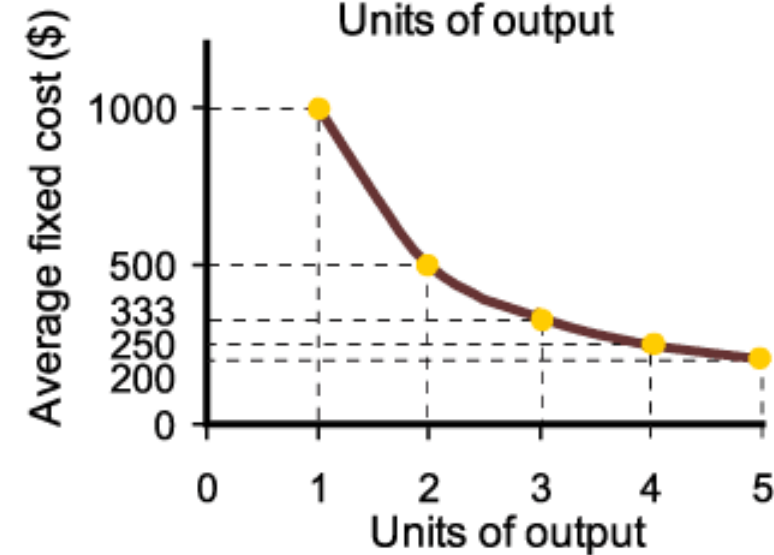
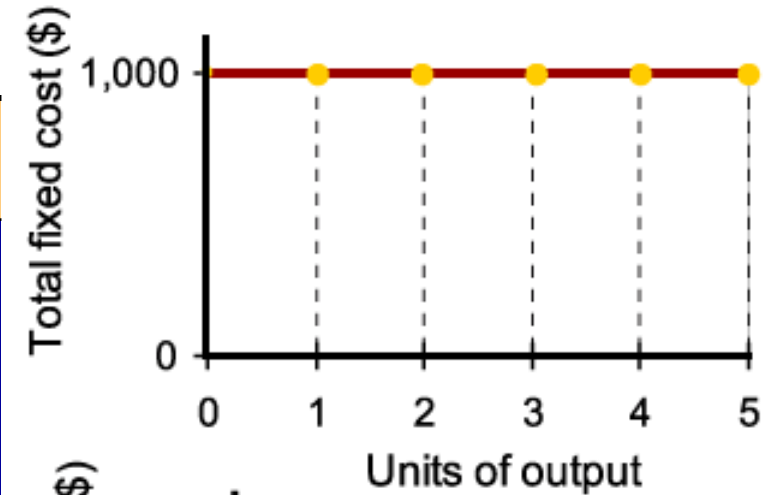
- Firms have no control over fixed costs in the short run. For this reason, fixed costs are sometimes called ***sunk costs***.
- **Average fixed cost (AFC)** is the total fixed cost (TFC) divided by the number of units of output (q):

$$AFC = \frac{TFC}{q}$$

Short-Run Fixed Cost (Total and Average) of a Hypothetical Firm

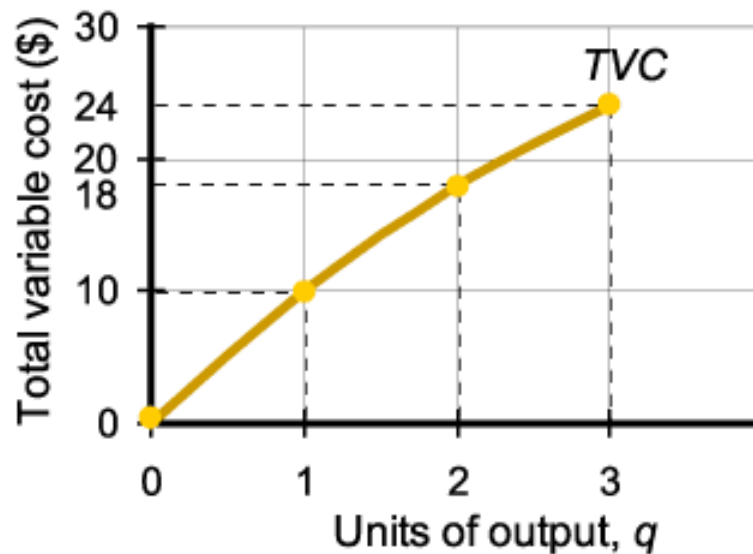
(1) q	(2) TFC	(3) AFC (TFC/q)
0	\$1,000	\$ —
1	1,000	1,000
2	1,000	500
3	1,000	333
4	1,000	250
5	1,000	200

- AFC falls as output rises; a phenomenon sometimes called **spreading overhead**.



Variable Costs

- The ***total variable cost curve*** is a graph that shows the relationship between total variable cost and the level of a firm's output.



- The total variable cost is derived from production requirements and input prices.

Derivation of Total Variable Cost Schedule from Technology and Factor Prices

PRODUCT	USING TECHNIQUE	UNITS OF INPUT REQUIRED (PRODUCTION FUNCTION)		TOTAL VARIABLE COST ASSUMING $P_K = \$2, P_L = \1 $TVC = (K \times P_K) + (L \times P_L)$
		K	L	
1 Units of output	A	4	4	$(4 \times \$2) + (4 \times \$1) = \$12$
	B	2	6	$(2 \times \$2) + (6 \times \$1) = \$10$
2 Units of output	A	7	6	$(7 \times \$2) + (6 \times \$1) = \$20$
	B	4	10	$(4 \times \$2) + (10 \times \$1) = \$18$
3 Units of output	A	9	6	$(9 \times \$2) + (6 \times \$1) = \$24$
	B	6	14	$(6 \times \$2) + (14 \times \$1) = \$26$

- The total variable cost curve shows the cost of production using the best available technique at each output level, given current factor prices.

Marginal Cost

- ***Marginal cost (MC)*** is the increase in total cost that results from producing one more unit of output.
- Marginal cost reflects changes in variable costs.

$$MC = \frac{\Delta TC}{\Delta Q} = \frac{\Delta TFC}{\Delta Q} + \frac{\Delta TVC}{\Delta Q}$$

Derivation of Marginal Cost from Total Variable Cost

UNITS OF OUTPUT	TOTAL VARIABLE COSTS (\$)	MARGINAL COSTS (\$)
0	0	0
1	10	10
2	18	8
3	24	6

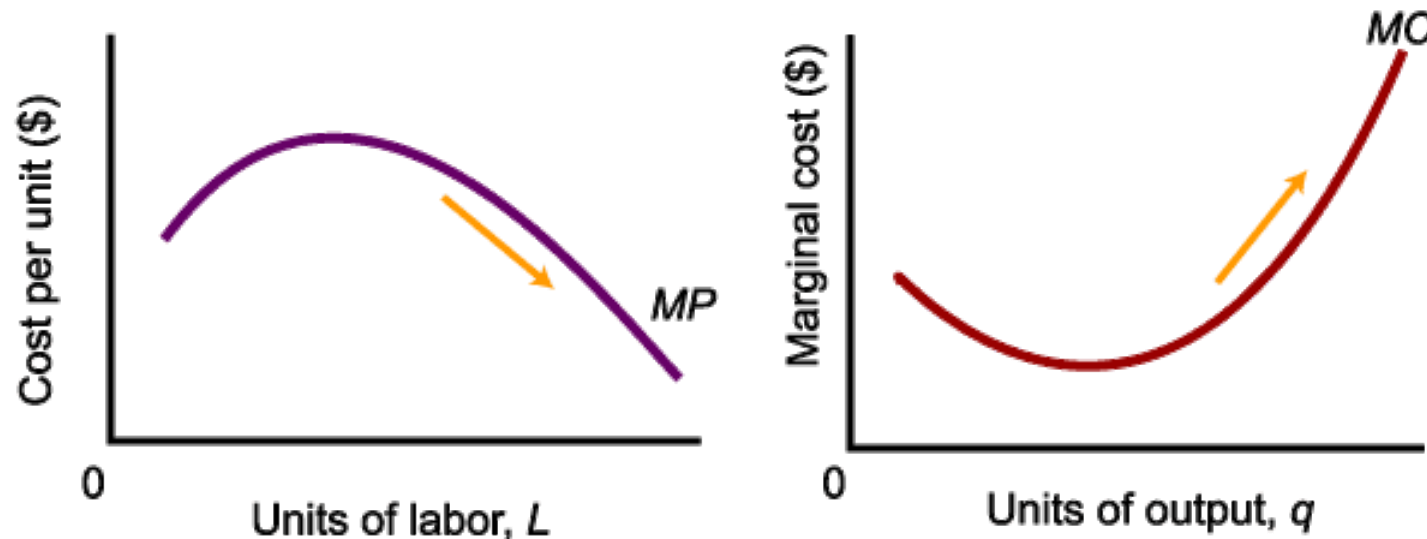
- **Marginal cost** measures the **additional** cost of inputs required to produce each successive unit of output.

The Shape of the Marginal Cost Curve in the Short Run

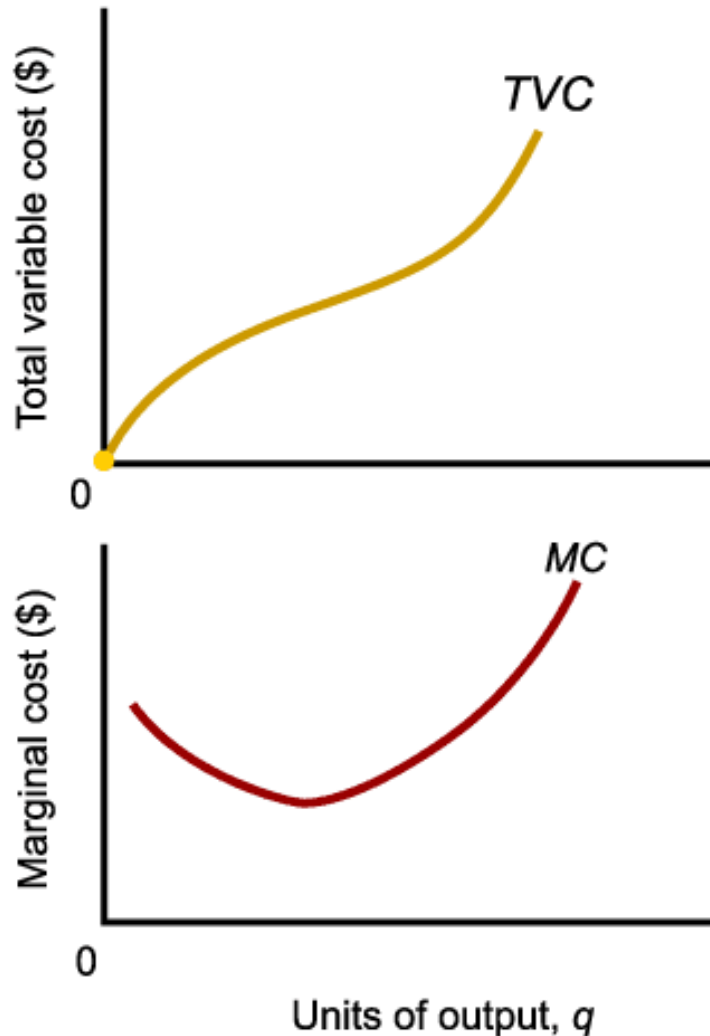
- The fact that in the short run every firm is constrained by some fixed input means that:
 1. The firm faces diminishing returns to variable inputs, and
 2. The firm has limited capacity to produce output.
- As a firm approaches that capacity, it becomes increasingly costly to produce successively higher levels of output.

The Shape of the Marginal Cost Curve in the Short Run

- Marginal costs ultimately increase with output in the short run.



Graphing Total Variable Costs and Marginal Costs



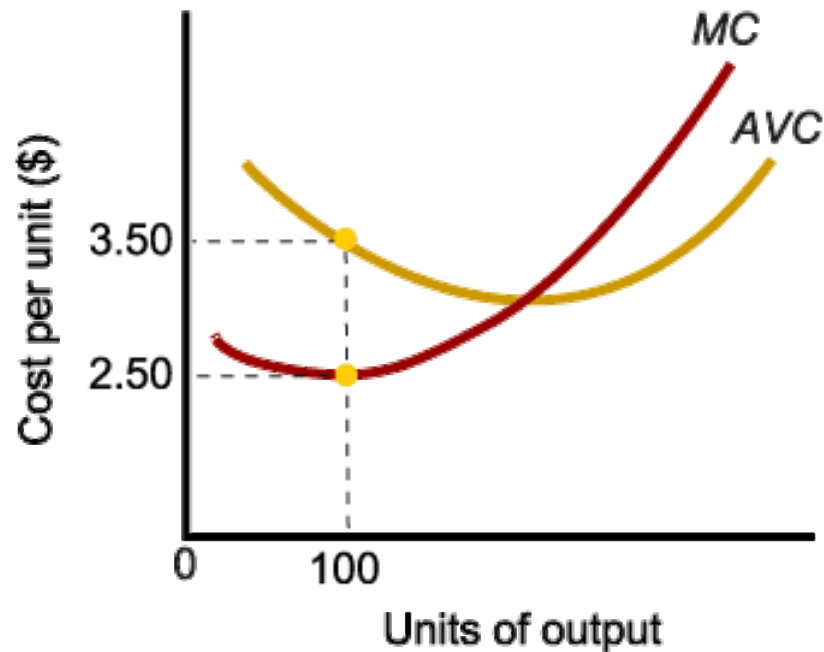
- Total variable costs always increase with output. The marginal cost curve shows how total variable cost changes with single unit increases in total output.
- Below 100 units of output, *TVC* increases at a *decreasing rate*. Beyond 100 units of output, *TVC* increases at an *increasing rate*.

- **Average Variable Cost (AVC)** is the total variable cost divided by the number of units of output:
$$AVC = \frac{TVC}{q}$$
- **Marginal cost (MC)** is the cost of producing *one additional unit*, whereas **average variable cost** is the variable cost spread across *all units* produced.
- **Average variable cost follows marginal cost**, but lags behind it.
- When $MC < AVC$, the average variable cost is falling.
- When $MC > AVC$, the average variable cost is rising.
- MC intersects AVC at its minimum point ($MC = AVC$).

Average Variable Cost

- *Average variable cost (AVC)* is the total variable cost divided by the number of units of output.
- Marginal cost is the cost of *one additional unit*. Average variable cost is the average variable cost per unit of *all the units* being produced.
- Average variable cost *follows* marginal cost, but lags behind.

Relationship Between Average Variable Cost and Marginal Cost

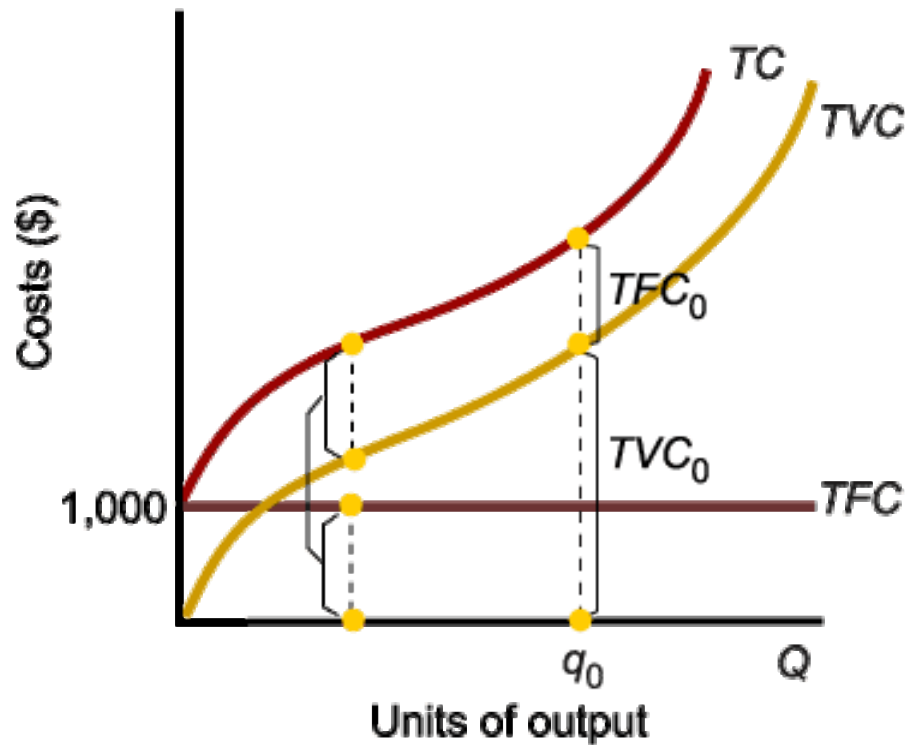


- When marginal cost is below average cost, average cost is declining.
- When marginal cost is above average cost, average cost is increasing.
- Rising marginal cost intersects average variable cost at the minimum point of *AVC*.
- At 200 units of output, *AVC* is minimum, and $MC = AVC$.

Short-Run Costs of a Hypothetical Firm

(1) q	(2) TVC	(3) MC (ΔTVC)	(4) AVC (TVC/q)	(5) TFC	(6) TC ($TVC + TFC$)	(7) AFC (TFC/q)	(8) ATC (TC/q or $AFC + AVC$)
0	\$ 0	\$ –	\$ –	\$1,000	\$ 1,000	\$ –	\$ –
1	10	10	10	1,000	1,010	1,000	1,010
2	18	8	9	1,000	1,018	500	509
3	24	6	8	1,000	1,024	333	341
4	32	8	8	1,000	1,032	250	258
5	42	10	8.4	1,000	1,042	200	208.4
–	–	–	–	–	–	–	–
–	–	–	–	–	–	–	–
–	–	–	–	–	–	–	–
500	8,000	20	16	1,000	9,000	2	18

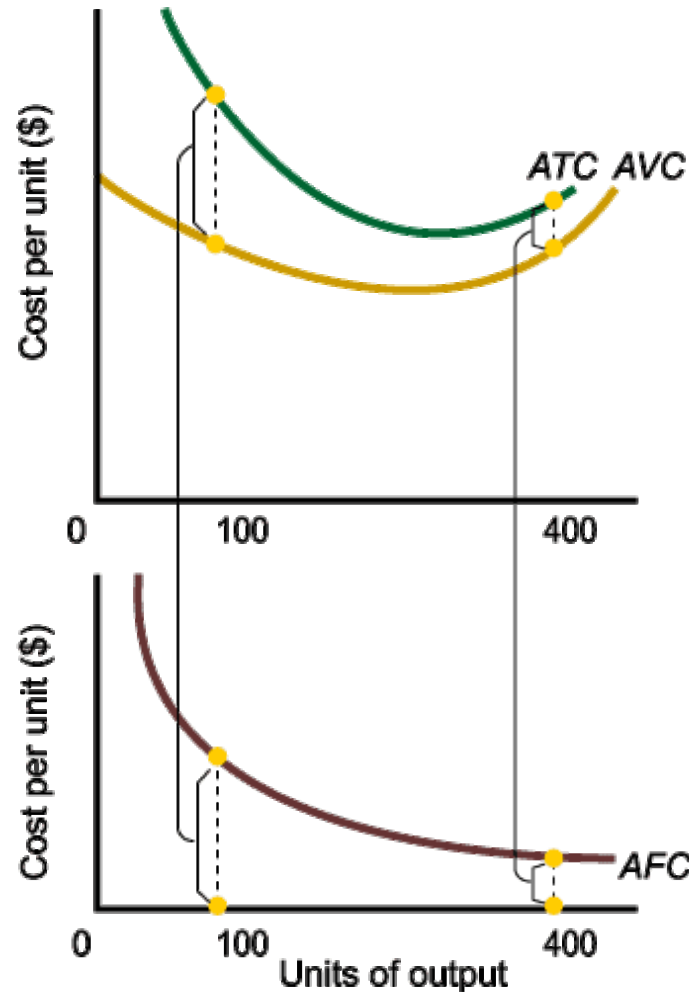
Total Costs



- Adding TFC to TVC means adding the same amount of total fixed cost to every level of total variable cost.
- Thus, the total cost curve has the same shape as the total variable cost curve; it is simply higher by an amount equal to TFC .

$$TC = TFC + TVC$$

Average Total Cost



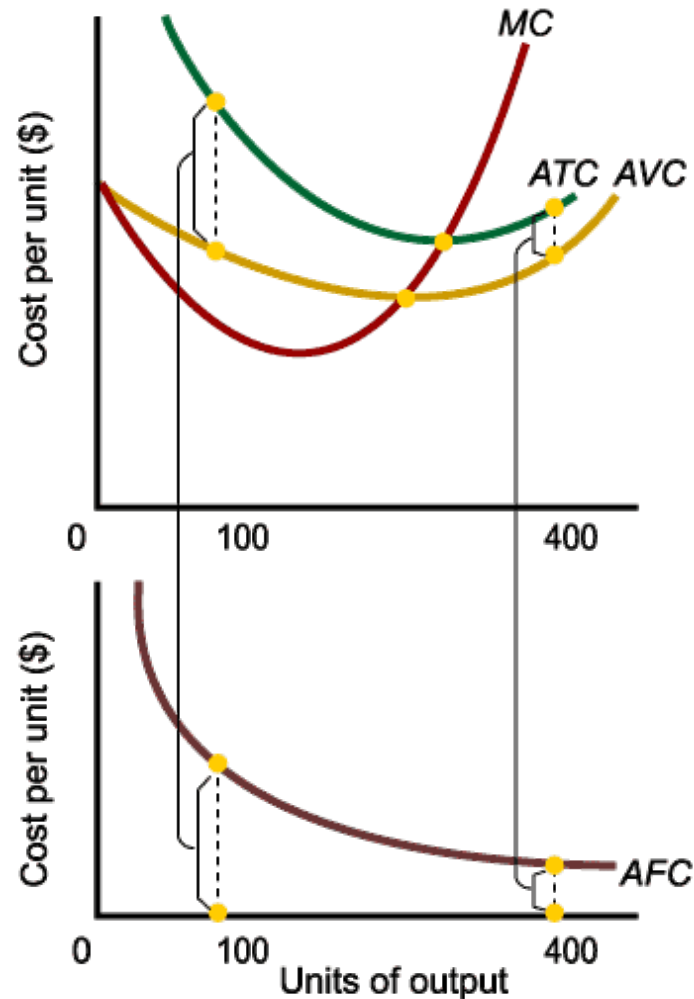
- Average total cost (ATC) is total cost divided by the number of units of output (q).

$$ATC = AFC + AVC$$

$$ATC = \frac{TC}{q} = \frac{TFC}{q} + \frac{TVC}{q}$$

- Because AFC falls with output, an ever-declining amount is added to AVC .

Relationship Between Average Total Cost and Marginal Cost



- If marginal cost is below average total cost, average total cost will decline toward marginal cost.
- If marginal cost is above average total cost, average total cost will increase.
- Marginal cost intersects average total cost and average variable cost curves at their respective minimum points.

Long-Run Cost Curves

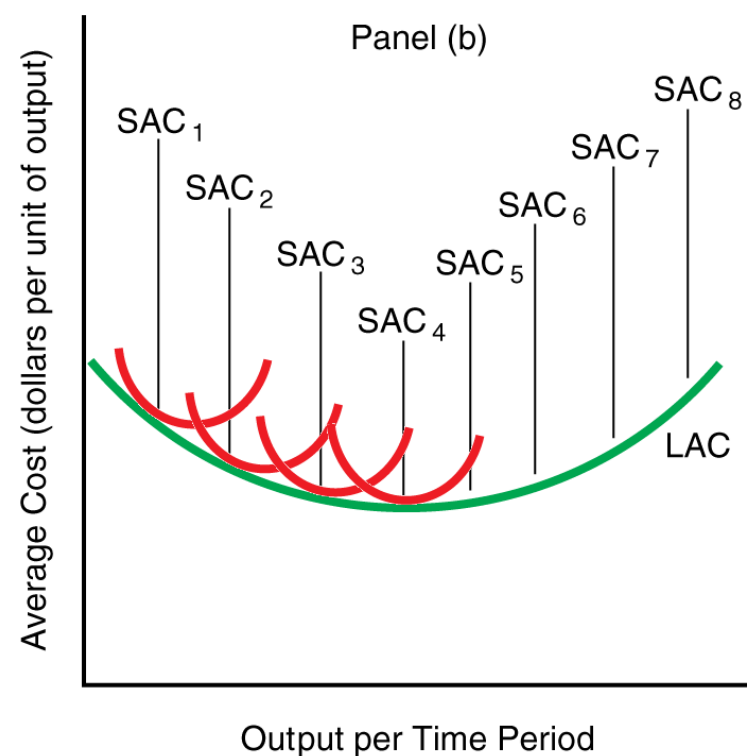
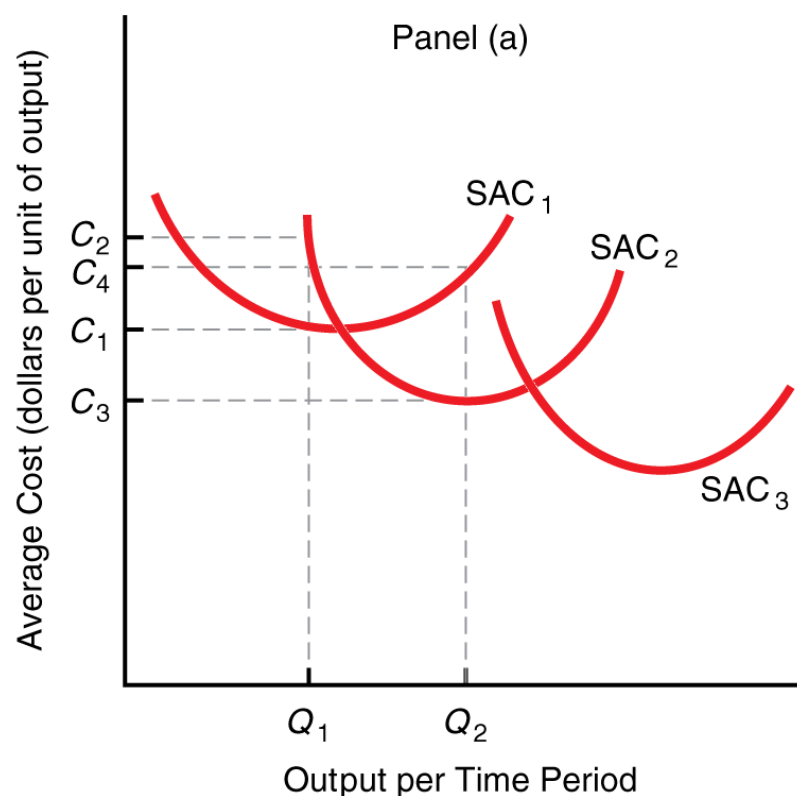
- Planning Horizon

- The long run, during which all inputs are variable

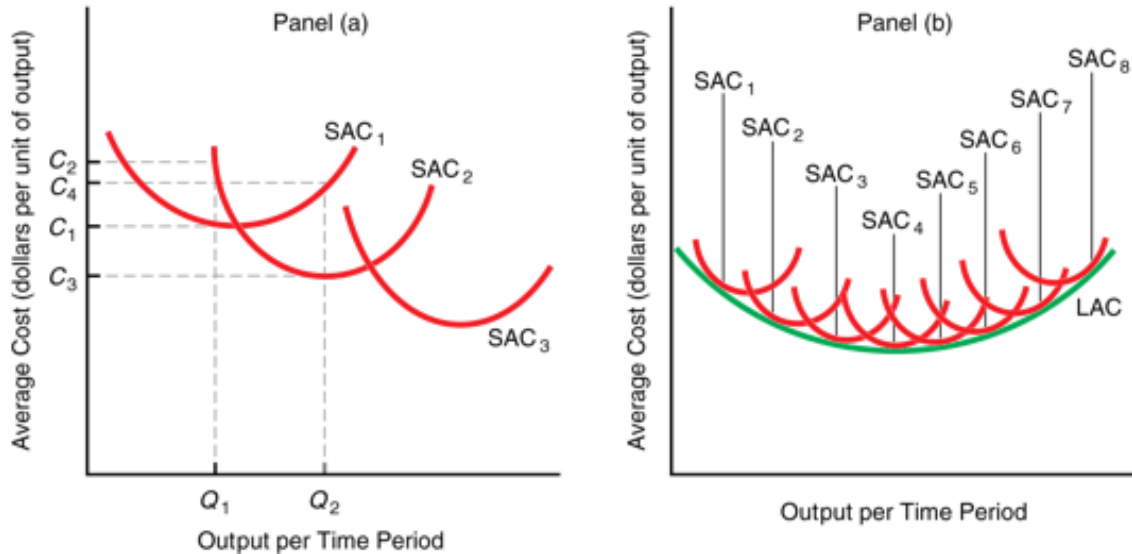
Preferable Plant Size and the Long-Run Average Cost Curve

Plant Size Selection

(Panel a): A firm selects the plant size (SAC_1, SAC_2, SAC_3) that yields the lowest average cost for its target output level (e.g., using SAC_1 for output Q_1 at cost C_1 , but shifting to SAC_2 for output Q_2 to lower unit costs from C_4 to C_3).



Preferable Plant Size and the Long-Run Average Cost Curve



- **Long-Run Flexibility:** In the long run, all inputs are variable, allowing the firm to expand or contract plant scale to achieve maximum cost efficiency.
- **Envelope Curve (Panel b):** The Long-Run Average Cost (*LAC*) curve envelopes the Short-Run Average Cost (*SAC*) curves, representing the lowest possible unit cost for producing any given output.

- **Continuous Choices:** As available plant options increase from discrete choices to an infinite series of scales, the *LAC* curve becomes a smooth curve tangent to each individual *SAC* curve.
- **Scale Economics:** The U-shape of the *LAC* curve reflects economies of scale (costs falling as plant size expands) followed by diseconomies of scale (costs rising as plant scale grows too large).

Long-Run Cost Curves

- Long-Run Average Cost Curve

- The locus of points representing the minimum unit cost of producing any given rate of output, given current technology and resource prices
- Only at minimum of long-run average cost curve, the short-run average cost curve is tangent to long-run average cost curve at their respective minimum

Why the Long-Run Average Cost Curve is U-Shaped

- Economies of scale
- Constant returns to scale
- Diseconomies of scale

Why the Long-Run Average Cost Curve is U-Shaped

- **Economies of Scale (Downward Slope):** As output expands initially, long-run average costs fall due to factors like specialization, bulk purchasing, and efficient technology.
- **Constant Returns to Scale (Flat Region):** At intermediate production levels, long-run average costs remain constant as output increases proportionally with input costs.
- **Diseconomies of Scale (Upward Slope):** Beyond a certain scale, expansion causes long-run average costs to rise due to management bottlenecks, coordination issues, and communication breakdown.
- **Overall U-Shape:** The transition across these three phases—falling costs, stable costs, and rising costs—gives the Long-Run Average Cost (*LAC*) curve its characteristic U-shape.

Application of Traditional Cost Curves

- **Predatory Pricing:** A aggressive pricing strategy where a dominant firm intentionally sets prices below its cost of production to eliminate competitors or create barriers for new entrants.
- Imposition of Rs 55.5 Crore on NSE by Competition Commission of India (CCI)
- $MC=MR$ is the basic principle of profit maximization.
- However, firms in the short run operate at prices below their Marginal Cost and also Average Total Cost (ATC) i.e. incurring losses. But at least they should earn AVC.
- If price is below the AVC even in the short run firm ceases to exist.
- **Predatory pricing case against Flipkart**

Other Cost Concepts

- **Sunk costs:** It is a past cost which cannot be altered by future action. So it is irrelevant but it is very difficult to ignore.
- **For e.g.** you bought 1000 shares at Rs. 25 per share and now it is priced at Rs. 15. There are other shares available which may have a better future than share presently possessed. But many people hold on to their present share until they recover their losses from same share.

Other Cost Concepts

- **Opportunity cost:** It is the cost of opportunity lost or foregone in terms of next best alternative.
- For e.g.: A given amount of money can be invested in any one of the alternatives:
 1. Shares/bonds
 2. Gold
 3. Real Estate
- You can choose an alternative which has least opportunity cost.

Other Cost Concepts

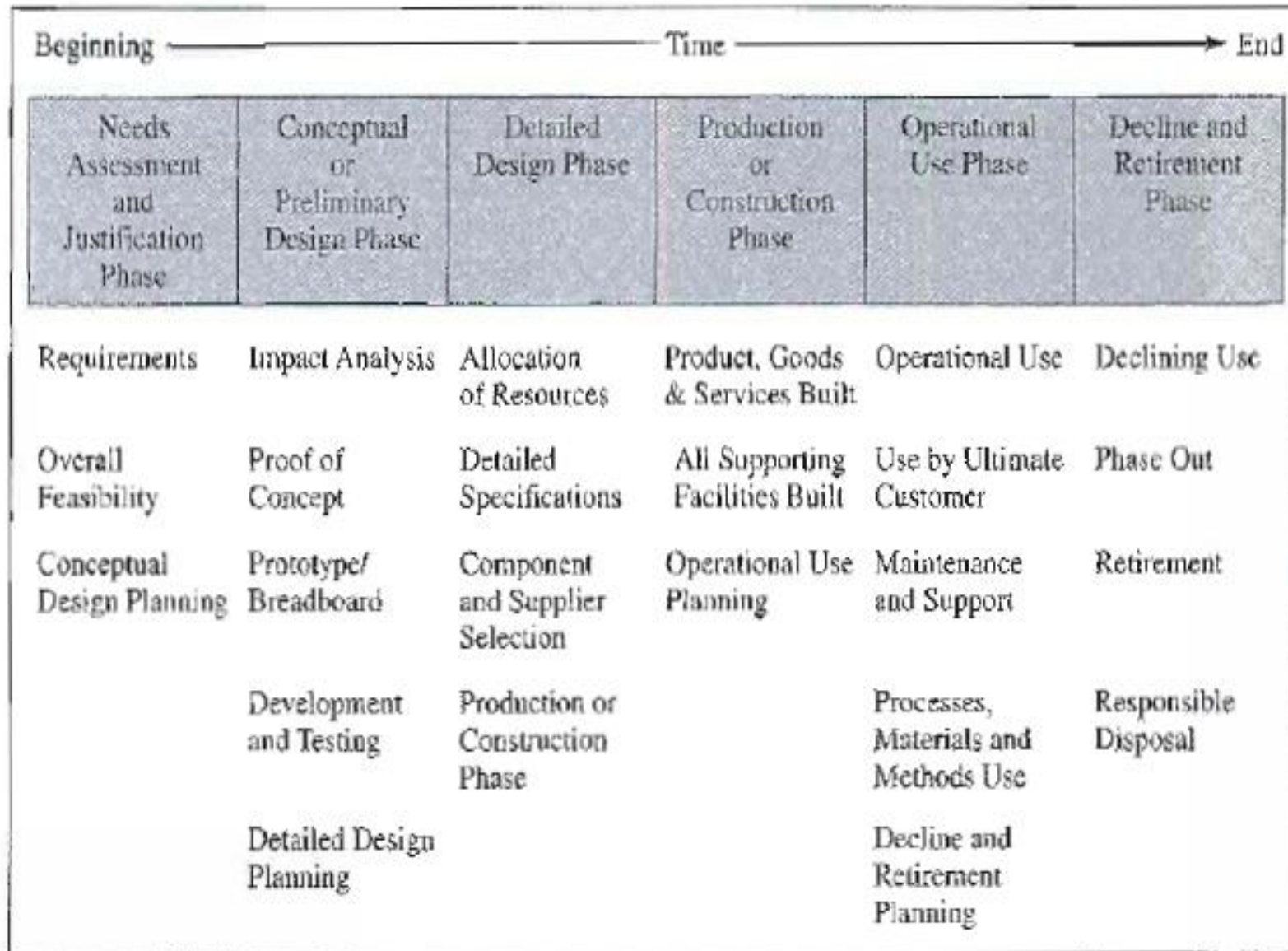
- **Recurring costs** refer to any expense that is known, anticipated, and occurs at regular intervals.
- **Nonrecurring costs** are one-of-a-kind expenses that occur at irregular intervals and thus are sometimes difficult to plan for or anticipate from a budgeting perspective

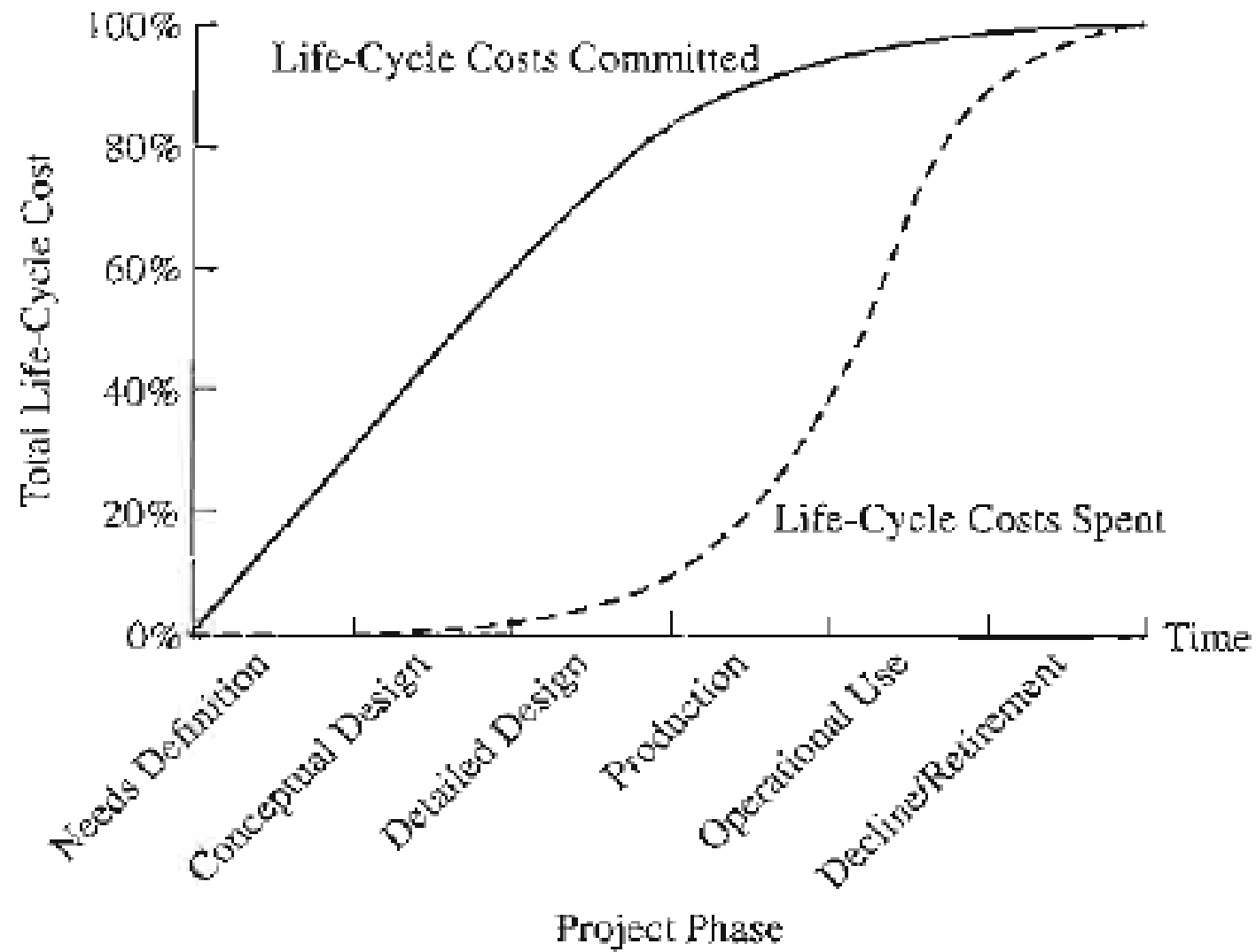
Other Cost Concepts

- **Incremental Costs:** One of the fundamental principles in engineering economic analysis is that in making a choice among a set of competing alternatives, focus should be placed on the *differences* between those alternatives
- **For e.g.:** There may be incremental costs associated with one option not required or stipulated by the other. In comparing the two leases, the focus should be on the differences between the alternatives, not on the costs that are the same.

Other Cost Concepts

- **Life Cycle Cost:** The products, goods, and services designed by engineers all progress through a life cycle.
- Life-cycle costing refers to the concept of designing products, goods, and services with a full and explicit recognition of the associated costs over the various phases of their life cycles.





- Figure illustrates how costs are committed early in the product life cycle-nearly 70-90% of all costs are set during the design phases. At the same time, as the figure shows, only 10-30% of cumulative life-cycle costs have been spent.
- Two key concepts in life-cycle costing are that the later design changes are made, the higher the costs, and that decisions made early in the life cycle tend to "lock in" costs that are incurred later.

Life-cycle design change costs and ease of change

